Parameter Estimation

Data Science

Parameter Estimation

- least squares
- maximum likelihood method
- maximum a posteriori
- confidence intervals

Statistical Inference

two main goals:

- **parameter estimation**: estimate unknown parameters from data
- **hypothesis testing**: make decisions about assumptions/models (discussed later in context of model evaluation)

What is Parameter Estimation?

assumption:

data generation process

observed data $\{\boldsymbol{x}_i\}_{i=1}^{n}$ sampled from a probability distribution of random variables $\boldsymbol{X}$

$$p_{\boldsymbol{X}}(\boldsymbol{x}|\boldsymbol{\theta})$$

→ model

with unknown parameters $\boldsymbol{\theta}$

goal: estimate $\boldsymbol{\theta}$ from observed data

Notation

mild notation overload
(to be handled by context)

- random variable: X
- vector of several random variables: $\boldsymbol{X}$

- observation of random variable X: x
- observation of random vector $\boldsymbol{X}$: $\boldsymbol{x}$
  - column vector: $\boldsymbol{x}$
  - row vector: $\boldsymbol{x}^{\mathrm{T}}$

features

- multiple observations
  - of X: $x_1, \cdots, x_n$ or $\{x_i\}_{i=1}^n$
  - of $\boldsymbol{X}$: $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n$ or $\{\boldsymbol{x}_i\}_{i=1}^n$

samples

- data matrix X
  - rows: samples
  - columns: features

- parameter: θ
- vector of parameters $\boldsymbol{\theta}$

- probability that X takes on value x: $p(x)$

probability distribution
abbreviation for $p_X(x)$

Two Types of Parameter Estimation Tasks

unsupervised setup (no labels)

- data: $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n$
- model: $p(\boldsymbol{x}; \boldsymbol{\theta})$
- goal: fit distribution of data

supervised setup (with labels)

- data: $(\boldsymbol{x}_i, y_i)$ pairs
- model: $p(y|\boldsymbol{x}; \boldsymbol{\theta})$
- goal: predict y from $\boldsymbol{x}$

ML perspective:

unsupervised learning
(e.g., GMM in clustering)

supervised learning

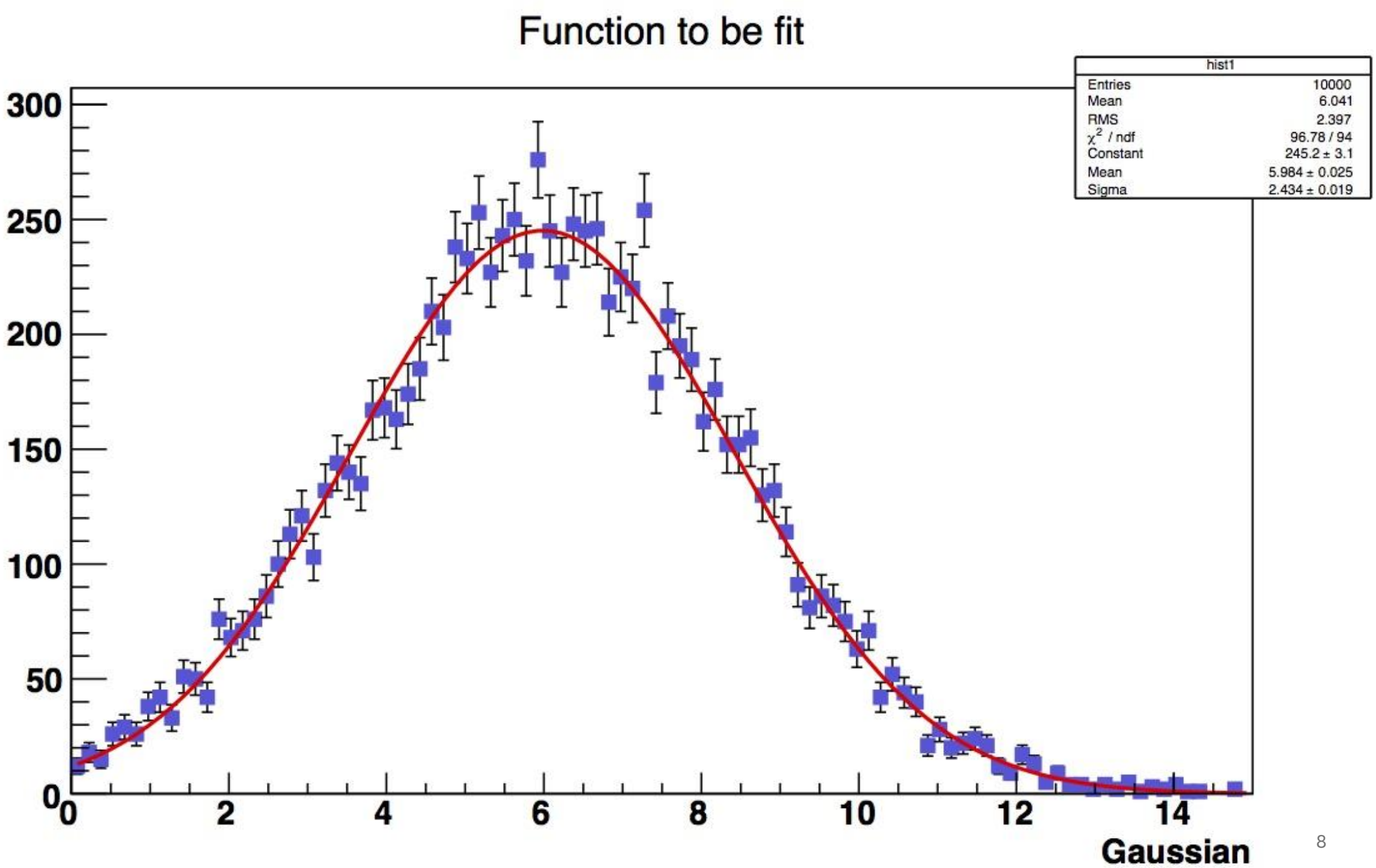
Unsupervised Setup: Examples

mean of a distribution of $X \sim p(x; \boldsymbol{\theta})$

- parameter of interest: $\mu = \mathbb{E}[X]$ (think of μ as θ_1)
- estimated from observed data $x_1, \cdots, x_n$: $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- examples: mean of Gaussian, probability of success in a Bernoulli

variance of a distribution (e.g., variance of Gaussian)

- parameter of interest: $\sigma^2 = \mathrm{Var}[X]$ (think of σ^2 as θ_2)
- estimated from sample variance

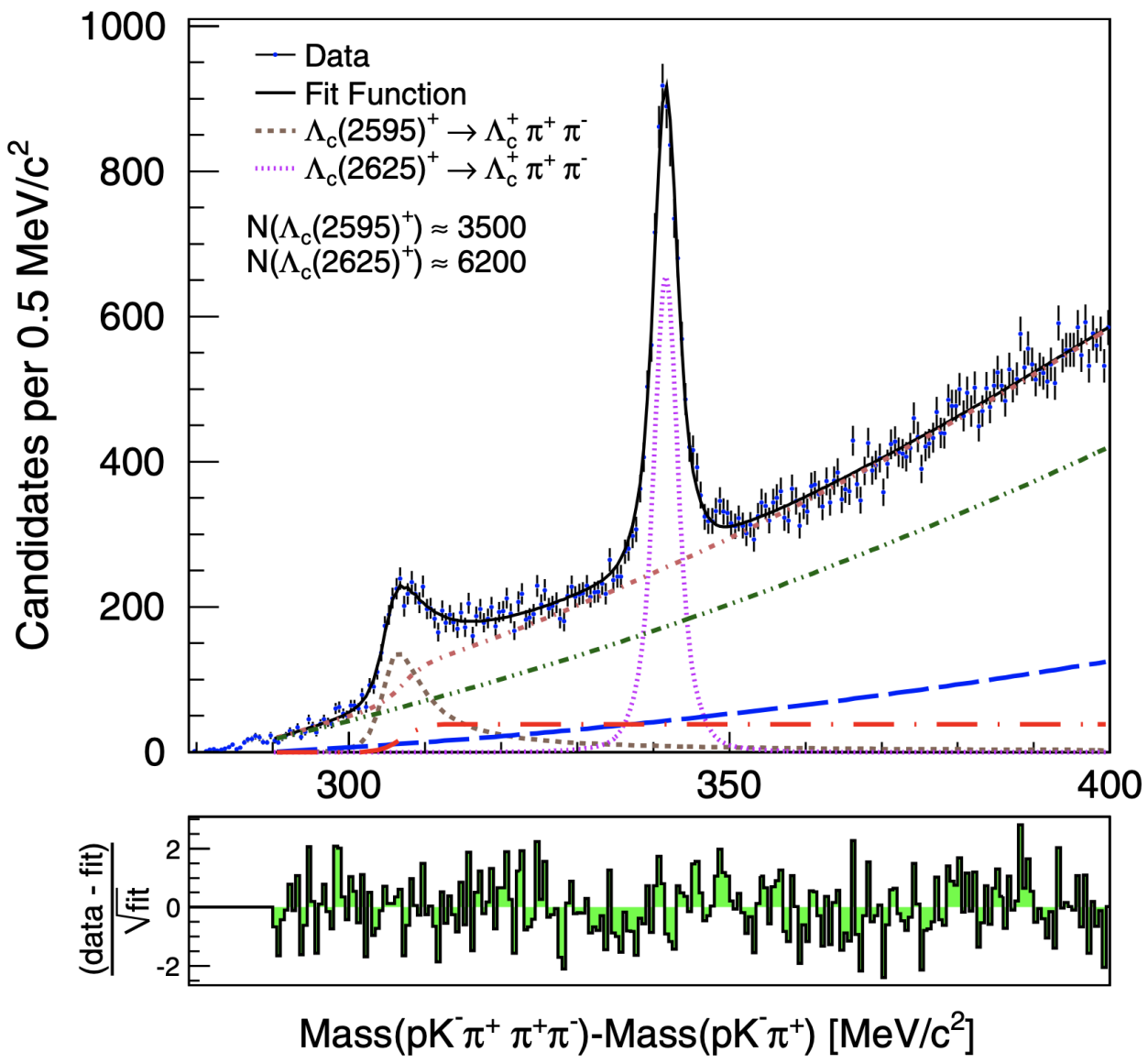

Function to be fit
hist1
Entries 10000
Mean 6.041
RMS 2.397
χ² / ndf 96.78 / 94
Constant 245.2 ± 3.1
Mean 5.984 ± 0.025
Sigma 2.434 ± 0.019
300
250
200
150
100
50
0
0
2
4
6
8
10
12
14
Gaussian

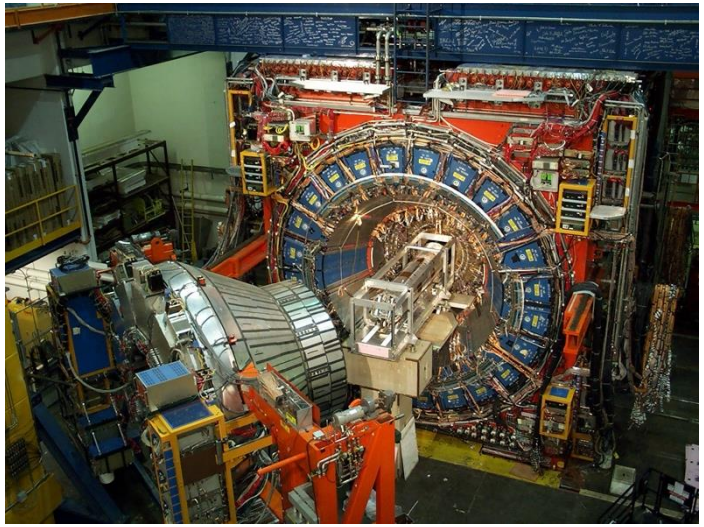
Unsupervised Setup: General Principle

estimate any θ of assumed $p(x; \boldsymbol{\theta})$

find parameter values such that the assumed model best explains the observed data

→ typically requires solving optimization problem (e.g. least squares, maximum likelihood)


Data
Fit Function
$\Lambda_c(2595)^+ \to \Lambda_c^+ \pi^+ \pi^-$
$\Lambda_c(2625)^+ \to \Lambda_c^+ \pi^+ \pi^-$
$N(\Lambda_c(2595)^+) \approx 3500$
$N(\Lambda_c(2625)^+) \approx 6200$
Candidates per 0.5 MeV/c^2
1000
800
600
400
200
0
300
350
400
$\frac{(data - fit)}{\sqrt{fit}}$
2
0
-2
Mass($pK^-\pi^+ \pi^+\pi^-$)-Mass($pK^-\pi^+$) [MeV/c^2]



Supervised Setup

$\boldsymbol{X} \sim p(\boldsymbol{x}), Y|\boldsymbol{X} = \boldsymbol{x} \sim p(y|\boldsymbol{x}; \boldsymbol{\theta})$

→ sampling conditional $(\boldsymbol{x}, y)$ pairs

goal: estimate $\boldsymbol{\theta}$ from observed data $(\boldsymbol{x}_1, y_1), \cdots, (\boldsymbol{x}_n, y_n)$ by fitting $f(\boldsymbol{x}_i; \boldsymbol{\theta})$ to y_i

→ later predict $\hat{y}$ for new $\boldsymbol{x}$ from learned $f\left(\boldsymbol{x}; \widehat{\boldsymbol{\theta}}\right)$

- regression parameters (continuous outputs $y \in \mathbb{R}$)
- classification parameters (discrete outputs $y \in \{1, \cdots, K\}$)

Example: Linear Regression

model (for each observation i):

$$y_i = \underbrace{\boldsymbol{w} \cdot \boldsymbol{x}_i + b}_{f(\boldsymbol{x}_i; \boldsymbol{w}, b)} + \varepsilon_i$$

typically several variables $\boldsymbol{X}$
→ also several parameters $\boldsymbol{w}$

predictive parameters $\boldsymbol{w}, b$
(σ^2 nuisance parameter)

assumption:

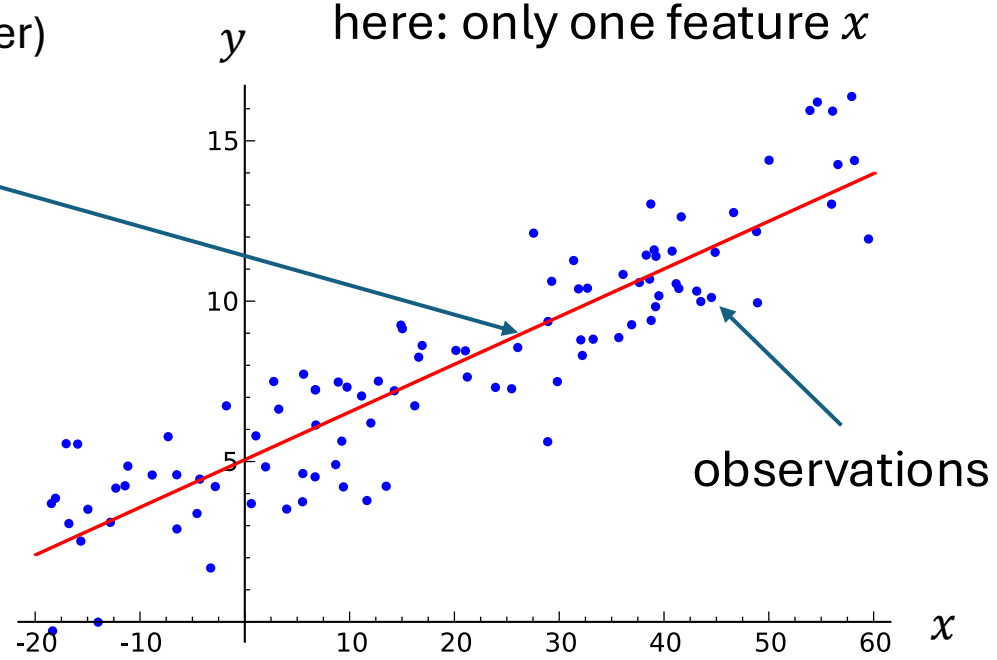
$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\rightarrow p(y|\boldsymbol{x}; w, b, \sigma^2) = \mathcal{N}(\boldsymbol{w} \cdot \boldsymbol{x} + b, \sigma^2)$$

different $p(y|\boldsymbol{x})$ for each $\boldsymbol{x}$,
sharing parameters $\underbrace{w, b, \sigma^2}_{\boldsymbol{\theta}}$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - f(\boldsymbol{x}_i; \boldsymbol{w}, b)\right)^2$$

here: only one feature x



observations

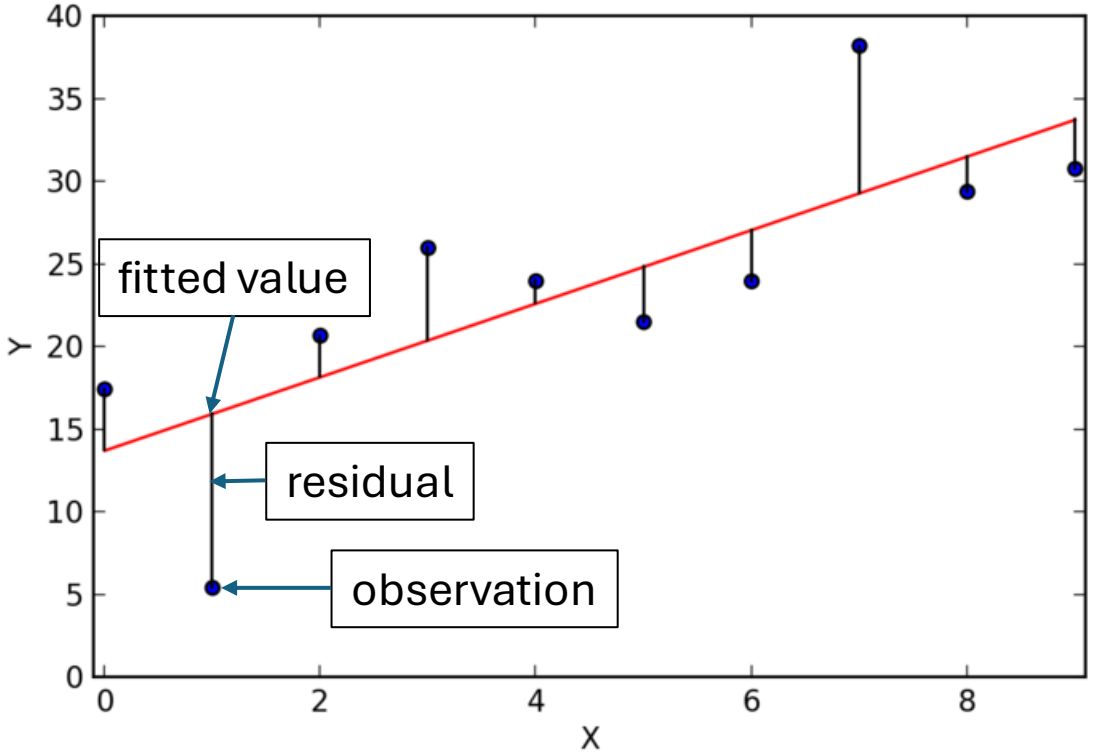
Least Squares Estimation

idea: choose parameters that minimize sum of squared residuals r_i

$$\underset{\boldsymbol{\theta}}{\operatorname{argmin}} \sum_{i=1}^{n} r_i(\boldsymbol{\theta})^2$$

examples for residuals:

- mean estimation: $r_i(\mu) = x_i - \mu$
- regression: $r_i(\boldsymbol{\theta}) = y_i - f(\boldsymbol{x}_i; \boldsymbol{\theta})$



Least Squares Estimation: Minimization

minimization of objective

$$J(\boldsymbol{\theta}) = \sum_{i=1}^{n} r_i(\boldsymbol{\theta})^2$$

candidate solution: need to ensure it is a minimum

first-order optimality condition: set gradient of objective to zero

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \sum_{i=1}^{n} \underbrace{2 r_i(\boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} r_i(\boldsymbol{\theta})}_{\text{residuals appear multiplied by their sensitivity}} = 0$$

$$\nabla_{\boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial}{\partial \theta_1} \\ \frac{\partial}{\partial \theta_2} \\ \vdots \end{bmatrix}$$

→ $\widehat{\boldsymbol{\theta}}$ satisfies $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = 0$

Linear Least Squares

model (linear in parameters):

$$r_i(\boldsymbol{\theta}) = y_i - \boldsymbol{x}_i^T \boldsymbol{\theta}$$

objective: $J(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^T \boldsymbol{\theta}\right)^2$

matrix form: $J(\boldsymbol{\theta}) = \|\boldsymbol{y} - X\boldsymbol{\theta}\|^2$

new notation: $\boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ $X = \begin{bmatrix} \boldsymbol{x}_1^T \\ \vdots \\ \boldsymbol{x}_n^T \end{bmatrix}$

linear regression:

$r_i(\boldsymbol{w}, b) = y_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b)$

define $\boldsymbol{\theta} = \begin{bmatrix} \boldsymbol{w} \\ b \end{bmatrix}$ and $\tilde{\boldsymbol{x}} = \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix}$

→ $r_i(\boldsymbol{\theta}) = y_i - \tilde{\boldsymbol{x}}_i^T \boldsymbol{\theta}$

Linear Least Squares: Solution

$$J(\boldsymbol{\theta}) = \|\boldsymbol{y} - X\boldsymbol{\theta}\|^2 = (\boldsymbol{y} - X\boldsymbol{\theta})^T(\boldsymbol{y} - X\boldsymbol{\theta}) = \boldsymbol{y}^T\boldsymbol{y} - \boldsymbol{y}^T X\boldsymbol{\theta} - \boldsymbol{\theta}^T X^T\boldsymbol{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}$$
$$= \boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{\theta}^T X^T\boldsymbol{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}$$

$$\boldsymbol{y}^T X\boldsymbol{\theta} = (\boldsymbol{\theta}^T X^T\boldsymbol{y})^T = \boldsymbol{\theta}^T X^T\boldsymbol{y}$$

take gradient w.r.t. $\boldsymbol{\theta}$:

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = -2X^T\boldsymbol{y} + 2X^T X\boldsymbol{\theta} = 2X^T(X\boldsymbol{\theta} - \boldsymbol{y})$$

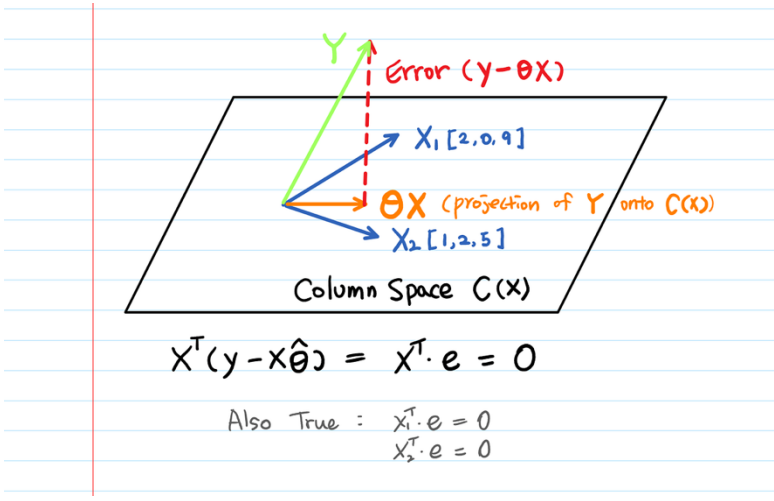
$\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^T A\boldsymbol{\theta}) = (A + A^T)\boldsymbol{\theta}$
here: $A = X^T X$, which is symmetric
→ $\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^T X^T X\boldsymbol{\theta}) = 2X^T X\boldsymbol{\theta}$

set gradient to zero (optimality condition):

$$X^T X\boldsymbol{\theta} = X^T\boldsymbol{y}$$

solution (normal equations):

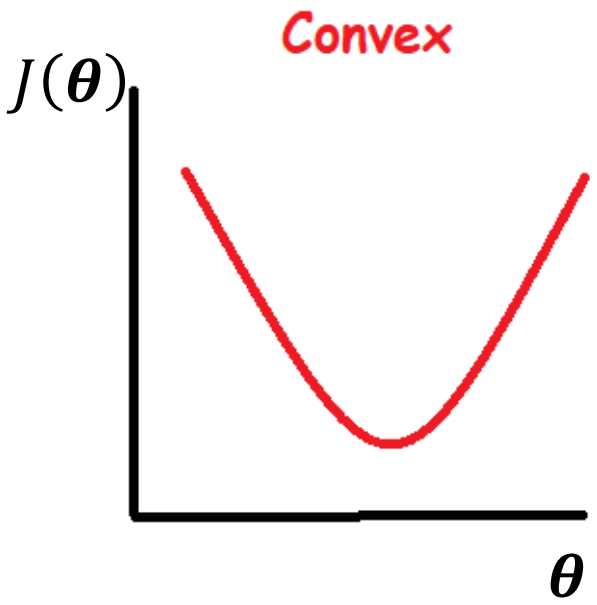
$$\widehat{\boldsymbol{\theta}} = (X^T X)^{-1} X^T\boldsymbol{y}$$

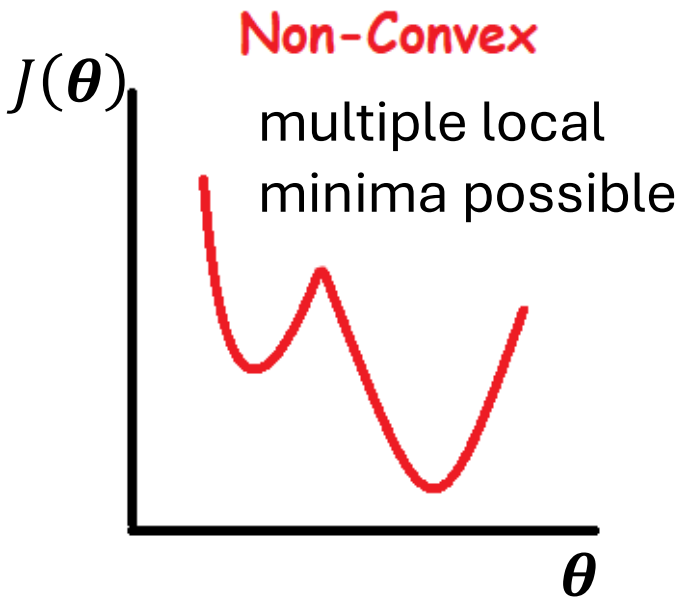


General Nonlinear Case

$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = 0$ gives stationary points

For convex problems (like linear least squares), this is the global minimum.





if $f(\boldsymbol{x}; \boldsymbol{\theta})$ nonlinear in $\boldsymbol{\theta}$: typically no closed-form solution to $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = 0$

→iterative numerical optimization

Gradient Descent

iteratively moving in direction of steepest descent (negative gradient):

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \cdot \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$$

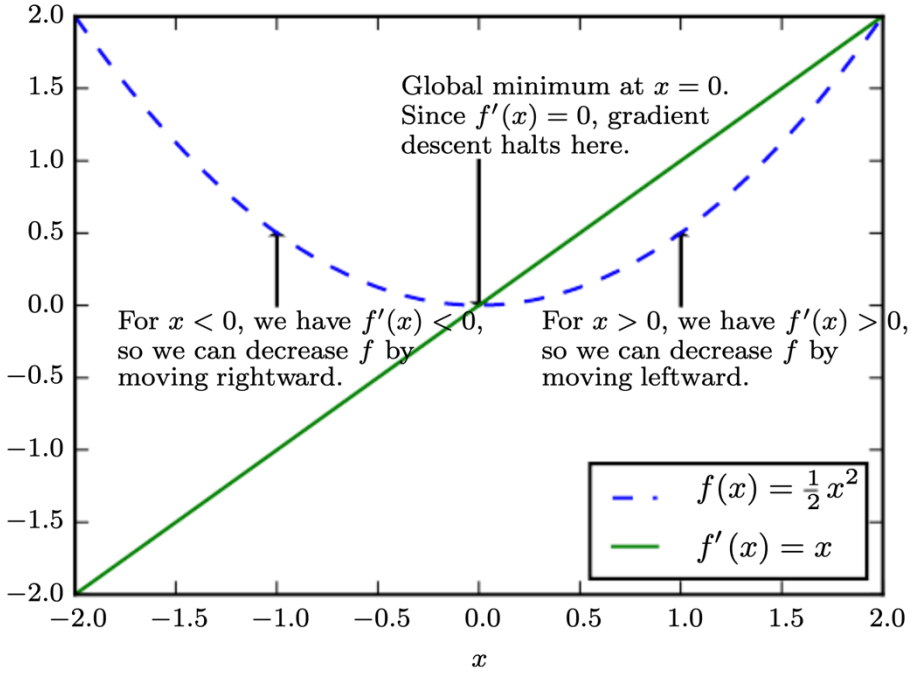
first-order

step size (learning rate): hyperparameter controlling how aggressively to move downhill

repeatedly take small steps downhill on the loss surface until a minimum is reached

typically use automatic differentiation (autodiff) to calculate gradients (systematic application of chain rule on computation graph)

called backpropagation in neural networks



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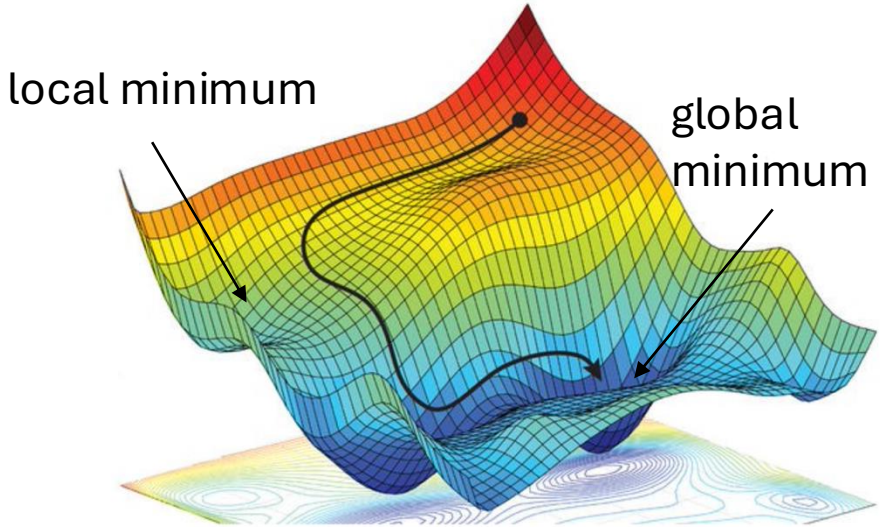


Beyond Gradient Descent

gradient descent is simple and general, but not always the best choice

alternative optimization methods:

- second-order methods → faster convergence, more expensive
  - use curvature (Hessian)
  - Newton's method, quasi-Newton (BFGS)
- stochastic methods (subsampling) → can escape shallow local minima, scales to large datasets, less stable
  - stochastic gradient descent (SGD)
  - mini-batch optimization

} more on this later in deep learning discussion

→ appropriate optimization strategy depends on structure of $J(\boldsymbol{\theta})$

Generalization: Loss Functions

least squares: minimizes sum of squared residuals between predictions and targets (independent of the model used, linear model → linear least squares)

loss functions generalize squared residuals to arbitrary prediction errors (still independent of the model used): $L_i(\boldsymbol{\theta}) = L\big(y_i, f(\boldsymbol{x}_i; \boldsymbol{\theta})\big)$

$L_i(\boldsymbol{\theta})$ quantifies prediction error (e.g., residual) on one sample

further examples:

- absolute error
- cross-entropy (classification)

Cost Function

aggregate (average) losses over data set:

$$J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} L_i(\boldsymbol{\theta})$$

to be minimized according to model parameters $\boldsymbol{\theta}$

→ objective function

L^2 Regularization (Ridge Regression)

add **penalty term** on parameter L^2 norm to cost function

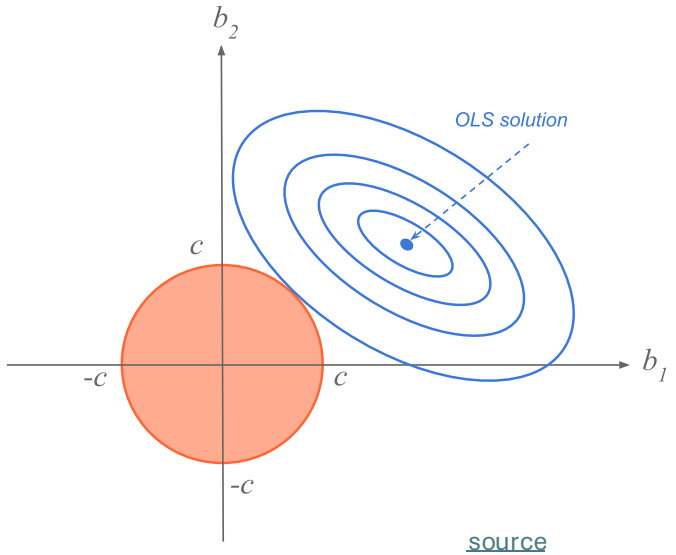
$$\tilde{J}(\boldsymbol{\theta}) = J(\boldsymbol{\theta}) + \lambda \cdot \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\theta}$$

hyperparameter controlling strength of regularization

aka **weight decay** in neural networks

corresponds to imposing constraint on parameters to lie in L^2 region (size controlled by λ)

goal: reduce overfitting from large/unstable parameters



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Maximum Likelihood Estimation (MLE)

assume data is generated by $p(y|\boldsymbol{x};\boldsymbol{\theta})$

idea: choose $\boldsymbol{\theta}$ to make observed data most likely

likelihood function

$$\widehat{\boldsymbol{\theta}} = \text{argmax}_{\boldsymbol{\theta}} \prod_{i=1}^{n} p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = \text{argmax}_{\boldsymbol{\theta}}\, \mathcal{L}(\boldsymbol{\theta})$$

independent and identically distributed (i.i.d.) samples

→ many loss functions defined from probabilistic assumptions

Side Note: i.i.d. Samples

identically distributed:

each sample drawn from the same distribution: $y_i \sim p(y|\boldsymbol{x}_i; \boldsymbol{\theta})$

→ same model, same parameters for all samples

independent:

→ additive losses: decompose over samples

samples do not influence each other: $p(y_1, \cdots, y_n|X) = \prod_{i=1}^{n} p(y_i|\boldsymbol{x}_i)$

→ knowing one observation gives no information about another

common violations: time series (temporal dependence), spatial data (spatial dependence), text/language (sequential dependence)

Negative Log-Likelihood

often written as log
(means ln, but changing the base
only introduces a constant factor)

log-likelihood function

$$\ell(\boldsymbol{\theta}) = \ln(\mathcal{L}(\boldsymbol{\theta})) = \sum_{i=1}^{n} \ln\big(p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})\big)$$

- logarithm monotonic function → maximum of ℓ and $\mathcal{L}$ at same $\boldsymbol{\theta}$ values
- most probability distributions (e.g., exponential family) only logarithmically concave (important for optimization)
- sum computationally more convenient than product

negative log-likelihood: minimization instead of maximization

$\widehat{\boldsymbol{\theta}} = \text{argmax}_{\boldsymbol{\theta}}\, \ell(\boldsymbol{\theta}) = \text{argmin}_{\boldsymbol{\theta}}\, \big(-\ell(\boldsymbol{\theta})\big)$ → $\nabla_{\boldsymbol{\theta}}\big(-\ell(\boldsymbol{\theta})\big) = 0$

→ negative log-likelihood: loss function derived from the likelihood

From Gaussian Noise to Least Squares

model assumption: $y_i = f(\boldsymbol{x}_i; \boldsymbol{\theta}) + \varepsilon_i \quad$ with $\quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$
implied distribution: $Y_i | X_i = x_i \sim \mathcal{N}(f(\boldsymbol{x}_i; \boldsymbol{\theta}), \sigma^2)$
likelihood:

$$p(y_i | \boldsymbol{x}_i; \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(- \frac{\left(y_i - f(\boldsymbol{x}_i; \boldsymbol{\theta})\right)^2}{2\sigma^2} \right)$$

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - f(\boldsymbol{x}_i; \boldsymbol{\theta})\right)^2 + \text{const}$$

$$\widehat{\boldsymbol{\theta}} = \text{argmax}_{\boldsymbol{\theta}}\, \ell(\boldsymbol{\theta}) = \text{argmin}_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left(y_i - f(\boldsymbol{x}_i; \boldsymbol{\theta})\right)^2$$

→ least squares corresponds to MLE assuming Gaussian noise

different noise assumptions → different loss functions
(other examples: Bernoulli → cross-entropy, Laplace → absolute error)

Two Views on Uncertainty

frequentist perspective

- parameters $\boldsymbol{\theta}$ fixed, but unknown
- data random: uncertainty comes from sampling variability
- estimate $\boldsymbol{\theta}$ using data (likelihood) only (point estimates, potentially with confidence intervals)

Bayesian perspective

- parameters as random variables, following $p(\boldsymbol{\theta}|y, \boldsymbol{x})$
- combine data (likelihood) and prior beliefs to express uncertainty via estimated posterior distributions

Bayes Theorem

H : hypothesis (model)
E : evidence (data)

likelihood function (derived from statistical model)

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

posterior probability (belief updated with observed evidence)

new data (same for all hypotheses → uninteresting)

prior probability (belief in H before evidence)

Bayes Theorem for Parameter Estimation

data likelihood:
model (function of $\boldsymbol{\theta}$ with data fixed)

$$p(\boldsymbol{\theta}|y,\boldsymbol{x}) \propto p(y|\boldsymbol{x},\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})$$

posterior:
probability distribution reflecting the effect of data on prior belief about parameters (increasing certainty with new evidence)

prior (e.g., broad Gaussian):
probability distribution reflecting initial belief about uncertain parameters (helps to reduce variance with few data)

to be compared to frequentist point estimates $\widehat{\boldsymbol{\theta}}$

→ Bayesian updating

Maximum A Posteriori (MAP) Estimation

MLE: mode of likelihood as point estimate

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MLE}} = \mathrm{argmax}_{\boldsymbol{\theta}}\, \mathcal{L}(\boldsymbol{\theta}) = \mathrm{argmax}_{\boldsymbol{\theta}} \prod_{i=1}^{n} p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$$

uses only the data, ignores prior knowledge

MAP: instead of only likelihood, include prior

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \mathrm{argmax}_{\boldsymbol{\theta}} \prod_{i=1}^{n} p(\boldsymbol{\theta}|y_i,\boldsymbol{x}_i) = \mathrm{argmax}_{\boldsymbol{\theta}} \prod_{i=1}^{n} p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})$$

likelihood (MLE)

prior (regularization)

Bayes

log: penalty term

- approximation to Bayesian inference (using mode of posterior distribution)
- MLE special case of MAP estimation with uniform prior

Gaussian Prior → Ridge Regression

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \mathrm{argmin}_{\boldsymbol{\theta}}\left[-\sum_{i=1}^{n}\log\left(p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})\right) - \log p(\boldsymbol{\theta})\right]$$

log of Gaussian prior $\boldsymbol{\theta}\sim\mathcal{N}(0,\tau^2 I)$:

$$\log p(\boldsymbol{\theta}) = -\frac{1}{2\tau^2}\|\boldsymbol{\theta}^2\| + \mathrm{const}$$

prior belief:
- parameters centered around zero
- large values unlikely

log of Gaussian likelihood:

$$\log\left(p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})\right) = -\frac{1}{2\sigma^2}\left(y_i - f(\boldsymbol{x}_i;\boldsymbol{\theta})\right)^2 + \mathrm{const}$$

$$\rightarrow \widehat{\boldsymbol{\theta}} = \mathrm{argmin}_{\boldsymbol{\theta}}\left[\sum_{i=1}^{n}\left(y_i - f(\boldsymbol{x}_i;\boldsymbol{\theta})\right)^2 + \underbrace{\frac{\sigma^2}{\tau^2}}_{\lambda}\|\boldsymbol{\theta}^2\|\right]$$

Priors as Regularizers

prior distribution: leveraging information that cannot be found in training data

MAP estimation explains regularization as incorporating prior knowledge into parameter estimation

many regularized estimation strategies can be interpreted as MAP

- L^2 regularization corresponds to MAP with Gaussian prior on parameters
- L^1 regularization corresponds to MAP with isotropic Laplace prior on parameters

MLE: data only

MAP: data + prior (regularization)

From Point Estimates to Uncertainty

two ways to express uncertainty

- Bayesian: credible intervals (probability about $\boldsymbol{\theta}$)
- frequentist: confidence intervals

go back to frequentist view: parameters fixed, uncertainty comes from data

so far point estimates for $\boldsymbol{\theta}$ (single best value)

often need for certainty of these estimates

→ instead of a single value, calculate interval of plausible values

Confidence Intervals (CI)

95% CI: containing the true parameter in 95% of repeated experiments

expressed with α: $100(1-\alpha)\%$

ingredients for CI:

1. estimator $\widehat{\boldsymbol{\theta}}$ (fixed)
2. its variability (standard error)
3. distribution (normal/asymptotic)

general form: $\widehat{\boldsymbol{\theta}} \pm (\text{critical value}) \times (\text{standard error})$

Standard Error (SE)

measures how much the estimate varies due to sampling:
If we repeated the experiment, how much would $\widehat{\boldsymbol{\theta}}$ fluctuate?

example: mean

$$\mathrm{SE}(\mu) = \frac{s}{\sqrt{n}}$$

s: sample standard deviation

n: number of samples

CI depends on data: more data (repetitions of experiment) → smaller SE → more precise estimate (narrower interval)

Critical Value

probabilistic correction factor from Student's t-distribution:

$$t_{n-1,1-\frac{\alpha}{2}}$$

- large n → t approaches standard normal
- small n → heavier tails → wider intervals

SE uses an estimate of the variance, but does not account for the uncertainty of that estimate itself

→ t-distribution accounts for the extra uncertainty

Bootstrapping

idea: treat observed dataset as proxy for the true distribution and simulate new datasets from it

procedure: aproximate the sampling distribution by resampling the data

- given data x $x_1, \cdots, x_n$: draw a bootstrap sample (with replacement)
- compute estimate $\widehat{\boldsymbol{\theta}}_b$
- repeat B times: $\widehat{\boldsymbol{\theta}}_1, \cdots, \widehat{\boldsymbol{\theta}}_B$
- use empirical quantiles for CI: $\left[\text{quantile}_{\alpha/2}, \text{quantile}_{1-\alpha/2}\right]$